

Logos Experiment 6 Report

Weight Tying + SwiGLU with 512-Token Context

Purpose: Test whether increasing the context length from 256 to 512 improves the best Logos model so far while keeping the token budget per training step constant.

1. Experiment Overview

This run builds on the strongest previous Logos variant: the Weight Tying + SwiGLU model with a 256-token context window. The new experiment keeps the same main architecture but doubles the sequence length to 512 tokens. To avoid giving the longer-context model more tokens per update, the batch size is reduced from 64 to 32, keeping tokens per step constant at 16,384.

The scientific question for this run is:

Does a longer context window improve validation loss and perplexity for the current Logos architecture under the same fixed training budget?

Component	Experiment 6 Setting
Run name	logos_exp06_weight_tying_swiglu_ctx512
Base architecture	Decoder-only causal Transformer
Tokenizer	GPT-2 tokenizer
Dataset	OpenWebText subset, 10,000 samples
FFN type	SwiGLU
Weight tying	Enabled
Context length	512 tokens
Batch size	32
Tokens per step	16,384
Training iterations	1,500

2. Controlled Change

The only intended experimental change is the context length. Experiment 5 used a 256-token block size with batch size 64. Experiment 6 uses a 512-token block size with batch size 32. Because both settings process 16,384 tokens per step, the comparison is cleaner than simply increasing context length while also increasing the token budget.

Variable	Experiment 5	Experiment 6
Context length	256	512
Batch size	64	32
Tokens per step	16,384	16,384
FFN type	SwiGLU	SwiGLU
Weight tying	Enabled	Enabled
Main hypothesis	Best current compact model	Longer context may capture longer dependencies

3. Final Results

Metric	Experiment 6 Result
Parameters	19,307,936
Best iteration	1,200
Best validation loss	5.8229
Final validation loss	5.8448
Final validation perplexity	345.43
Checkpoint size	73.69 MB
Peak VRAM	11.18 GB
Training time	157.20 sec, approximately 2.62 min
Training speed	Approximately 157,371 tokens/sec
Generation speed	73.43 tokens/sec
Generation latency	13.62 ms/token
Average GPU power	248.77 W
Estimated GPU energy	10.72 Wh

4. Comparison with Experiment 5

Experiment 5 is the most important comparison point because it uses the same Weight Tying + SwiGLU model but with a 256-token context. The table below shows that the 512-token version did not improve language modeling quality under the fixed training budget.

Metric	Exp 5: WT + SwiGLU, ctx 256	Exp 6: WT + SwiGLU, ctx 512	Better
Parameters	19.24M	19.31M	Exp 5
Context length	256	512	Exp 6
Batch size	64	32	Neutral
Tokens/step	16,384	16,384	Equal
Best val loss	5.7230	5.8229	Exp 5
Final val loss	5.7247	5.8448	Exp 5
Final perplexity	306.36	345.43	Exp 5
Checkpoint size	73.44 MB	73.69 MB	Exp 5
Peak VRAM	11.18 GB	11.18 GB	Equal
Training time	157.07 sec	157.20 sec	Similar
Generation latency	15.04 ms/token	13.62 ms/token	Exp 6
Energy	10.49 Wh	10.72 Wh	Exp 5

5. Interpretation

The 512-token context experiment did not improve validation loss or perplexity. In fact, the model became worse on the main quality metrics despite using the same number of tokens per training step. The final validation perplexity increased from 306.36 in Experiment 5 to 345.43 in Experiment 6.

This suggests that simply increasing context length is not automatically useful for the current small-data, short-training setup. Longer context can help language models, but it usually requires more training tokens, longer optimization, and enough long-range structure in the dataset for the model to benefit.

Observed tradeoffs:

- Quality decreased: validation loss and perplexity became worse than Experiment 5.
- Parameter count increased slightly because the positional embedding table doubled in length.
- Checkpoint size increased slightly from 73.44 MB to 73.69 MB.
- Peak VRAM stayed approximately the same at 11.18 GB because the batch size was reduced.
- Generation latency improved compared with Experiment 5, but the quality loss is more important for this run.
- Training speed and total training time remained very similar because tokens per step were kept constant.

6. Scientific Conclusion

Conclusion: Increasing the context length from 256 to 512 did not improve the current Logos model under the same fixed-budget training setup. Experiment 5 remains the best overall model so far because it achieved lower validation loss, lower perplexity, smaller checkpoint size, and lower energy use.

The likely reason is that the training budget was too small for the model to learn useful long-range dependencies. A longer context gives the model more possible information, but it also makes the optimization problem harder. If the model does not receive enough training steps and data, the extra context may not translate into better prediction quality.

7. Current Experiment Ranking

Rank	Experiment	Main Finding
1	Experiment 5: Weight Tying + SwiGLU, ctx 256	Best overall quality-per-parameter and quality-per-storage so far.
2	Experiment 4: SwiGLU FFN	Improved validation quality and generation latency at baseline parameter scale.
3	Experiment 1: Weight Tying	Reduced parameter count and checkpoint size by about 40 percent with almost no quality loss.
4	Baseline	Useful fixed-budget control experiment.
5	Experiment 6: Context 512	Longer context did not improve quality under the same budget.
6	Experiment 3: RMSNorm	Slightly worse quality and slower than LayerNorm baseline.
7	Experiment 2: GELU	Worse than ReLU under the same fixed budget.

8. Recommended Next Step

The best model so far remains Experiment 5: Weight Tying + SwiGLU with 256-token context. The next experiment should either improve this model directly or test whether its advantage holds under a longer training budget.

Option	Experiment	Reason
A	Train Experiment 5 longer	Check whether the best model continues improving with more iterations.
B	Reduce dropout in Experiment 5	The model may be underfitting in short runs.
C	Weight Tying + SwiGLU + better data filtering	Test data efficiency without increasing model size.
D	Weight Tying + SwiGLU + learning-rate tuning	The best architecture may improve further with tuned optimization.

9. One-Line Summary

Experiment 6 showed that doubling context length from 256 to 512 did not improve Logos under the same fixed training budget; the compact 256-context Weight Tying + SwiGLU model remains the strongest result so far.